Spectral Compression of Decimal Mountain Concatenations: Four Infinite Multiplicative-Order Families Beyond the Smooth Case

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Abstract

For $M(k) = 1 \parallel 2 \parallel \cdots \parallel k \parallel (k-1) \parallel \cdots \parallel 1$, we prove that every prime p with $\mathrm{ord}_p(10) = 7 \cdot 2^a 5^c$ or $\mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$, $a, c \geq 0$, $0 \leq b \leq 2$, divides at least one $M(k)$. The argument first compresses the stable affine drift for a *fixed* coprime-order component v to at most $\ell = \mathrm{lcm}(v, \mathrm{ord}_v(10))$ cyclotomic cosets, uniformly over arbitrarily large powers of $2, 5$. Both drift coordinates become polynomials of degree at most two in $m = r/\gcd(r, \ell)$ over a fixed-degree cyclotomic field. Exact quadratic ($v = 7$) and degenerate linear ($v = 11$) resultants, followed by factorization-free cyclotomic sieves, exclude simultaneous zero drift in these families. A nine-torsion lifting lemma folds the new cases $v = 33, 99$ into the same 22-phase algebra as $v = 11$, reusing its norms. This does not resolve the unrestricted OEIS A077187 conjecture.

1 Affine drift and the period reduction

For a prime $p \nmid 10$, set $r = \mathrm{ord}_p(10) = uv$ with $u = 2^a 5^c$, $(v, 10) = 1$. Write $C_k = 1 \parallel \cdots \parallel k$, $Y_k = 10^{-D(k)}(k \parallel \cdots \parallel 1) \pmod p$, where $D(k)$ is the total digit count. For a complete d-digit layer, put

$$L = 10^{d-1}, \quad q = 10^d, \quad n = 9L, \quad A = q^n, \quad S = \sum_{i<n} q^i, \quad J = \sum_{i<n} iq^i.$$

The exact layer map is

$$(C, Y) \longmapsto \big(AC + (q-1)S - J,\ A^{-1}(Y + LS + J)\big).$$

The established stable-period lemma gives, for

$$T = \mathrm{lcm}(r, \mathrm{ord}_v(10)), \qquad \mathrm{ord}_1(10) = 1,$$

a translation from the full block of digit layers $T+1, \ldots, 2T$. Let (U, V) denote its displacement. If $(U, V) \neq (0, 0)$, the normalized residues of $M(10^{jT})$, $M(10^{jT} + 1)$ have affine slopes $10U + V$ and $1000U + V$. For $p \geq 7$, $p \neq 11$, these cannot both vanish, yielding a divisible mountain with $1 \leq j \leq p$.

Set $h = \mathrm{ord}_v(10)$,

$$\ell = \mathrm{lcm}(v, h), \qquad g = \gcd(r, \ell), \qquad m = r/g, \qquad T = m\ell.$$

The key point is that ℓ depends only on v, not on a, c.

Lemma 1 (Short-period phase)**.** *For every stable digit layer d, the diagonal multiplier $A_d = 10^{9d10^{d-1}}$ has period ℓ. Furthermore $\prod_{j=d}^{d+\ell-1} A_j = 1 \pmod p$. Thus its cumulative phase P_d, started at $d = T + 1$, also has period ℓ.*

Proof. For stable d, $u \mid 9d10^{d-1}$. Modulo v, periodicity follows from $v \mid \ell$ and $10^\ell \equiv 1 \pmod v$. For the product, use

$$9 \sum_{j=d}^{d+\ell-1} j10^{j-1} = (d + \ell - 1)10^{d+\ell-1} - d10^{d-1} - \sum_{j=d}^{d+\ell-2} 10^j.$$

The exponent is zero modulo u termwise and modulo v because $v \mid \ell$ and $v \mid (10^\ell - 1)/9$; the latter follows from the lifting-the-exponent identity at 3 and invertibility of 9 at the remaining prime divisors. Therefore r divides the full exponent. □

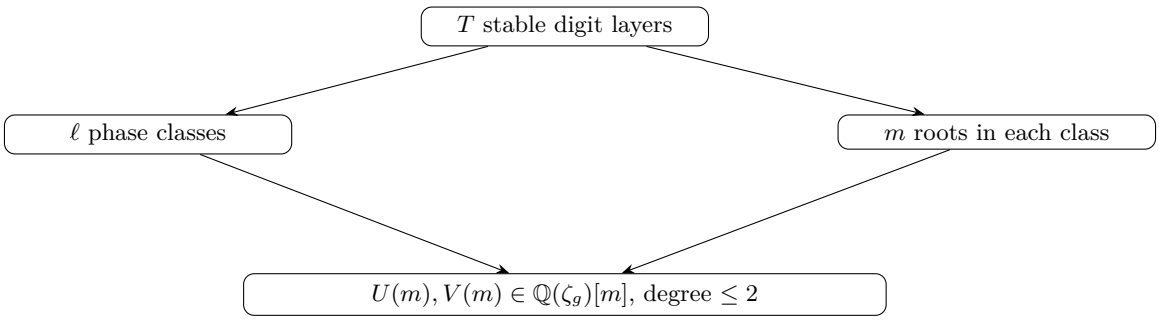


Figure 1: The unbounded stable block compresses to bounded-dimensional cyclotomic algebra once v is fixed.

2 Coset sums and quadratic spectral coordinates

Let $\zeta = 10^m \pmod p$, a primitive g-th root. For $j = 1, \ldots, \ell$, all layers $d = T + j + t\ell$ $(0 \leq t < m)$ share the same $A = A_j$ and $P = P_j$. Their values $q = 10^d$ are precisely the m roots of $z^m = c = \zeta^j$. Define

$$F_1 = \sum_{\substack{z^m = c \\ z \neq 1}} \frac{z}{z-1}, \qquad F_2 = \sum_{\substack{z^m = c \\ z \neq 1}} \frac{z}{(z-1)^2}.$$

Logarithmic differentiation of $z^m - c$, with $z = 1$ removed if necessary, gives

	F_1	F_2
$c \neq 1$	$mc/(c-1)$	$m^2c/(c-1)^2$
$c = 1$	$(m-1)/2$	$-(m^2-1)/12$.

For a nonsingular full layer, its ascending and descending block residues are

$$B = (A-1)\big(1 + q/(q-1)^2\big) - 9LA/(q-1),$$
$$R^* = (A-1)\big(L/(q-1) - q/(q-1)^2\big) + 9LA/(q-1).$$

The singular layer $q = 1$ contributes $B = R^* = 9/200$. Put $\alpha = P^{-1}(1 - A^{-1})$, $\beta = P(A-1)$, $n_0 = m - \mathbf{1}_{c=1}$.

Theorem 1 (Bounded-degree spectral compression)**.** *For each fixed v, the drift equals the sum of ℓ coset contributions*

$$\begin{aligned} U_j &= \alpha(n_0 + F_2) - \tfrac{9}{10}P^{-1}F_1 + \mathbf{1}_{c=1}\tfrac{9}{200}P^{-1}, \\ V_j &= \beta(F_1/10 - F_2) + \tfrac{9}{10}PAF_1 + \mathbf{1}_{c=1}\tfrac{9}{200}P. \end{aligned} \tag{1}$$

Moreover,

$$A_j = \zeta^{(g/v)\gamma\, 9j10^{j-1}}, \quad P_j = \prod_{t<j} A_t, \quad \gamma = u^{-1} \pmod v,$$

so U, V are polynomials of degree at most two in m, with coefficients in the fixed cyclotomic field $\mathbb{Q}(\zeta_g)$, where $g \leq \ell$.

Proof. In the ascending long word the j-th layer block has weight $(PA)^{-1}$, while the descending block has weight P. Substitute the two block residues, sum over the m roots in each phase class, and include the singular contribution. The displayed formulas follow. Their only dependence on m is through m, m^2, while all phases are powers of ζ_g. □

Lemma 2 (Involutive degree drop for $\mathrm{ord}_v(10) \leq 2$)**.** *If $v \mid 99$, then the coefficients of m^2 in both compressed drift coordinates vanish identically: $U_2 = V_2 = 0$ in the cyclotomic coefficient field.*

Proof. For $v = 1$, all stable multipliers equal 1. Otherwise $10^2 \equiv 1 \bmod v$ implies $A_{-j} = A_j^{-1}$. Because $\prod_{j=1}^{\ell} A_j = 1$, this yields the palindrome $P_{\ell+1-j} = P_j$. Hence both phase differences $\alpha_j = P_j^{-1} - P_{j+1}^{-1}$ and $\beta_j = P_{j+1} - P_j$ change sign under $j \mapsto \ell - j$. The coefficient of m^2 in F_2 is the inversion-even function $f(c) = c/(c-1)^2$ for $c \neq 1$, extended by $f(1) = -1/12$. Thus $U_2 = \sum_j \alpha_j f(\zeta^j)$ and $V_2 = -\sum_j \beta_j f(\zeta^j)$ cancel pairwise under the involution. □

The result also applies to $v = 33, 99$, although their linear resultant exclusions have not been established.

3 Two complete infinite order families

Theorem 2 (Exact computer-assisted families)**.** *Every prime satisfying*

$$\mathrm{ord}_p(10) \in \{7 \cdot 2^a 5^c,\ 11 \cdot 2^a 5^c : a, c \geq 0\}$$

divides at least one decimal mountain concatenation $M(k)$.

Proof. For $v = 7$, $h = 6$, $\ell = 42$, and $g = 7$ if $a = 0$, otherwise $g = 14$. The exact coordinates $U(m), V(m)$ are quadratic in m. Their Sylvester resultant

$$\mathcal{R} = (U_2V_0 - U_0V_2)^2 - (U_2V_1 - U_1V_2)(U_1V_0 - U_0V_1)$$

has, after scaling by 40000, a nonzero exact norm, independent of the six phase values γ:

g	$\lvert\mathrm{Norm}(40000\mathcal{R})\rvert$
7	33476324853960868807109434619064591936
14	11497454360840474493325194486198045857885568 9

For $v = 11$, $h = 2, \ell = 22$, and $g = 11$ or 22. Here the exact quadratic coefficients satisfy $U_2 = V_2 = 0$ in all ten phases, and we use the *linear* resultant $\mathcal{R} = U_1V_0 - U_0V_1$. All twenty scaled norms $\lvert\mathrm{Norm}(4000\mathcal{R})\rvert$ are nonzero and listed in the accompanying certificate.

If simultaneous zero drift occurred in either family, its prime p would divide the corresponding fixed nonzero norm N, satisfy $r = \mathrm{ord}_p(10) \mid p - 1$ and $p \mid \Phi_r(10)$, hence $r < p \leq N$. The exact scripts evaluate $\Phi_r(10) \bmod N$ using base cyclotomic polynomials of degree at most 24 or 40 and modular powering. For all $53 + 4479$ admissible indices in the $v = 7$ norms, and all 3943 phase-dependent indices in the $v = 11$ norms, they verify that $\gcd(N, \Phi_r(10))$ has no prime divisor outside r, by repeated gcd removal. Since a prime of exact order r cannot divide r, zero drift is impossible. The affine two-offset witness theorem finishes the proof. □

4 Nine-torsion lifting: two more infinite families

Lemma 3 (Nine-torsion folding)**.** *Let v_0 be coprime to 30, and put $\ell_0 = \mathrm{lcm}(v_0, \mathrm{ord}_{v_0}(10))$, assuming $3 \nmid \ell_0$. For $s \in \{1, 3, 9\}$, suppose $\mathrm{ord}_p(10) = r = sv_0u$, $u = 2^a5^c$, and let $g_0 = \gcd(r, \ell_0)$, $n = r/g_0$, $\xi = 10^n \bmod p$. The stable multipliers and cumulative phases are ℓ_0-periodic, and the full drift is expressed using the* same *fixed-v_0 spectral formulas as in (1), with $m = n$ and phase $\gamma' = (su)^{-1} \bmod v_0$.*

Proof. Because $s \mid 9$, the stable multiplier exponents $9d10^{d-1}$ vanish modulo s. Their shift by ℓ_0 vanishes modulo v_0 since $v_0 \mid \ell_0$ and $10^{\ell_0} \equiv 1 \bmod v_0$; it vanishes modulo u in the stable range. The product of ℓ_0 consecutive multipliers is 1: the exponent is divisible by u and s, and modulo v_0 this follows from $v_0 \mid \ell_0$ and $v_0 \mid (10^{\ell_0} - 1)/9$. Therefore P_d has the same short period. Now $T = \mathrm{lcm}(r, \ell_0) = n\ell_0$. For each $j \leq \ell_0$, the n values $q_{T+j+t\ell_0}$ are exactly the roots of $z^n = \xi^j$, while division of the stable exponent by $n = su/(g_0/v_0)$ gives

$$A_{T+j} = \xi^{(g_0/v_0)\gamma' 9j10^{j-1}}.$$

Substituting into the universal coset sums (1) proves the claim, including the singular root. □

Corollary 1 (Infinite families with $v = 33, 99$)**.** *Every prime p with*

$$\boxed{\mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c, \quad a, c \geq 0, \quad b = 1, 2}$$

divides a decimal mountain $M(k)$.

Proof. Take $v_0 = 11$, so $\ell_0 = 22$. For $s = 3, 9$, the folding lemma reduces both drift coordinates to the already certified *linear* 22-phase polynomials, evaluated at $n = 11su/\gcd(r, 22)$ and $\gamma' = (su)^{-1} \bmod 11$. A zero drift would force p to divide the corresponding nonzero integer norm $N_{g_0,\gamma'} = \lvert\mathrm{Norm}(4000\mathcal{R})\rvert$ from the $v = 11$ case. Thus $r < p \leq N_{g_0,\gamma'}$ and $p \mid \Phi_r(10)$. The separate exact modular cyclotomic checker tests all 3,850 phase-admissible indices for $s = 3$ and all 3,757 for $s = 9$. It verifies

$$\gcd(N_{g_0,\gamma'}, \Phi_r(10)) = 1$$

in every case, using the same fixed twenty nonzero norms, without factoring them. This excludes zero drift for every exponent pair. The two-offset witness theorem finishes the proof. □

Lemma 4 (Exact 3-adic boundary)**.** *Suppose* $r = \mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$ *with* $b \geq 1$*, and set* $A_d = 10^{9d10^{d-1}}$ *on stable digit layers. For* $t \geq 0$*,*

$$\boxed{\mathrm{ord}(A_{d+22\cdot 3^t}/A_d) = 3^{\max(b-2-t,0)}.}$$

Consequently the 22-period folding is valid precisely for $b \leq 2$ *among these 3-power extensions.*

Proof. Let $L = 22 \cdot 3^t$ and $e_d = 9d10^{d-1}$. The exact difference

$$e_{d+L} - e_d = 9 \cdot 10^{d-1}\big(L + (d+L)(10^L - 1)\big)$$

is divisible by 11 and by the smooth factor $2^a 5^c$ in the stable range. By the 3-adic lifting-the-exponent lemma, $v_3(10^L - 1) = 2 + t$, while $v_3(L) = t$. Therefore $v_3(e_{d+L} - e_d) = 2 + t$ exactly. The claimed order follows from $\mathrm{ord}_p(10) = 11 \cdot 3^b 2^a 5^c$. □

The lifting lemma is algebraic; it does not automatically give an existence theorem for arbitrary v_0 or $s > 9$.

5 Reproducibility and scope

The files `certify_fixed_v7.py` and `certify_fixed_v11.py` generate all exact norms and check every phase-dependent cyclotomic gcd, without factoring the large norm integers or enumerating target primes. Independent prime-field tests are in `audit_fixed_v7.py` and `audit_fixed_v11.py`. The nine-torsion lift is independently checked by `certify_fixed_v33_v99.py` and `audit_fixed_v33_v99.py`. Dedicated CI jobs reproduce all certificates.

The spectral theorem applies to arbitrary fixed v, but a nonzero resultant and the elimination of its actual-order prime factors are *not* established for arbitrary v. The general all-primes A077187 problem remains open.

References

[1] OEIS Foundation, *A077187*, https://oeis.org/A077187.

[2] Q. Wang, *Decimal mountain concatenation research notes and exact certificates*, https://github.com/dongxuelian2/oeis-papers/pull/8, 2026.